```
    vectorizer = TfidfVectorizer(stop_words='english') tfidf_matrix =
vectorizer.fit_transform(sentences) scores = cosine_similarity(tfidf_matrix, tfidf_matrix) return
scores

# Step 3: Rank Sentences
def rank_sentences(scores, sentences): ranked = np.argsort(np.sum(scores, axis=1))[-5:][::-1] # Top 5
sentences summary = [sentences[i] for i in ranked]
    return " ".join(summary)

# Example Usage
if __name__ == "__main__":
    text = """
    Artificial intelligence and machine learning are transforming industries worldwide. They enable
businesses to make data-driven decisions and automate processes. These technologies are also
advancing fields like healthcare, education, and transportation. However, challenges such as ethical
concerns, bias, and data privacy remain crucial areas of focus.
    """

    sentences = preprocess_text(text)
    scores = calculate_tfidf(sentences)
    summary = rank_sentences(scores, sentences) print("Summary:")
    print(summary)
```

Project 6: Recommendation System for E-Commerce

Objective:

To develop a recommendation engine that suggests products based on user behavior, improving user experience and increasing sales conversions. The system will leverage collaborative and content-based filtering, enhanced by advanced machine learning techniques for personalized recommendations.

Detailed Plan:

1. Data Collection:
    - Gather user and product data from e-commerce platforms, including:
        - Purchase history, clickstream data, and product ratings.
        - Product attributes such as categories, price, and brand.